

Power over Metabolic Fire: Minimal Models of Affect, Purpose and Norms

David Balduzzi

Abstract. This paper argues that purpose and norms arise when learning is coupled to self-preservation. It develops minimal, differentiable models in which metabolism scales network activity, and sustaining metabolism is the objective. Two hypotheses connect the models to experience: that Jacobians, the first-order structure of physical interactions, characterize the structure of experience; and that affect is the experience of empowering or diminishing experience. Four consequences follow. (1) Coupling learning to metabolism yields a self-referential mechanism where weight updates rescale the system’s own experience; by the second hypothesis, that is affect. (2) When metabolism depends on the system’s actions, metabolic sensitivity assigns every parameter a preferred direction, set by the system’s ongoing need to endure, which gives goals and norms an intrinsic, physical foundation. (3) Modeling energy as a reserve replenished by past actions isolates a *metabolic channel*: accumulated past experience, time-decayed, reweighted by what each moment contributed to the reserve, and rescaled by how depleted the system is now. (4) Treating metabolism as a constraint rather than an objective recovers the same term as a KKT constraint gradient, gated by a multiplier that vanishes while energy is comfortable. Finally, the paper closes on ethics. Goal and norm creation, born of power over metabolic fire, is the fundamental capability separating organisms from today’s neural nets; giving it to them would be a mistake.

Code: github.com/dbalduzzi/gradland

1 Introduction

“no one has hitherto laid down the limits to the powers of the body, that is, no one has as yet been taught by experience what the body can accomplish solely by the laws of nature” —Spinoza [1]

What is special about brains and datacenters compared to almost everything else? How are they similar, and different, in form and function? The paper tackles these questions, at a high level, using two main tools. The first is differential functionalism, the bridge between interactions and experience developed

in Gradland [2]. The second is feedback, which has a long and remarkable history.

You can draw a straight line from steam engines to deep learning, connecting dots every four score years [3]. In 1698, twenty years after Spinoza’s *Ethics* and a decade after Newton’s *Principia*, Savery invented the first commercial steam pump. In 1788, Watt applied the fly-ball governor to steam engines, using negative feedback to maintain a steady speed across varying loads, turning steam engines into a *general purpose technology* with wide applications in factories [4]. Maxwell created control theory in 1868 – shortly after unifying electricity, magnetism and optics – by applying differential equations to characterize how the dynamics of governors are stabilized [5].

Rosenblueth, Wiener and Bigelow published the founding document of cybernetics “Behavior, Purpose and Teleology” [6] in 1943. They proposed that purpose is goal-directed behavior (behavior that may be interpreted as aimed at the attainment of a final condition) and teleology is purpose combined with negative feedback for error correction. Roughly, their definitions of purpose and teleology map to inference and learning (in-context learning complicates the picture). Wiener gave the name ‘cybernetics’ to a new science aimed at studying control in animal *and* machine. The name is an homage to Watt’s governor, using the same Greek root [7]. Today, learning-as-feedback is the conceptual foundation of modern machine learning, rebranded as artificial intelligence.

To think effectively about capabilities, and in particular their limits, requires a new perspective. Claims about task performance are a dead end. Modern AI is based on machine learning; learning is based on feedback. I draw on feedback, its role in the origin of cybernetics [6], and Jonas’ critique of that role [3], to understand how brains and datacenters differ in their experience and capabilities today – and to start thinking about some choices that lie ahead.

1.1 Motivation

There is an understandable tendency to anthropomorphize language models because they speak our language [8]. There is a lot of speculation about how intelligent they are, or could be; how autonomous

they are, or could be; and increasing curiosity about whether they are conscious [9–11]. Intelligence, autonomy, and consciousness are complex concepts that combine many factors.

The paper studies experience and feedback in organisms and AIs. Experience and feedback are more basic concepts than intelligence, agency, and consciousness, and are directly relevant to them.

1.2 Overview and related work

For good order, recall the main assumption from [2]

Hypothesis G. The first-order structure of physical interactions, i.e. Jacobians, characterizes the structure of experience and vice versa.

§2 reviews the varieties of experience encountered in [2], generic experience and the experience of learning, and gives a preview of the focus of this paper: the experience of sustaining. §3 describes complications that arise from the gap between abstract models and the reality of modern hardware. It highlights some architectural and structural differences between datacenters and brains.

§4 shifts to *function*, and specifically the role of metabolism. There is a substantial literature arguing that metabolism and interoception are fundamental to affect, purposiveness and norms, tracing back at least to Hans Jonas [12], see [13–28]. Lisa Feldman Barrett and others have provided ample empirical evidence linking metabolic regulation and interoception to emotion [29–35].

What is new is constructing minimal models of metabolism in the language of modern machine learning and the direct link to experience. For the direct link we make a second assumption: roughly, Hypothesis A is that *affect is the experience of elevating or degrading your experience, your self*. The hypothesis is unpacked in §4.2; it is painfully simple. The philosophy here, as in Gradland, is to let the models do the work. The main contribution of the paper is then to unfold the consequences of Hypothesis A when combined with minimal models of metabolism.

The models are developed in four steps, progressively adding structure and sharpening the implications for experience. First, metabolism as a factor scaling activity, §4.3. Coupled with learning, this yields a self-referential mechanism whereby weight updates rescale the system’s own experience. Combined with Hypothesis A, it immediately follows that metabolism is fundamental to affect. Second, metabolism made dependent on the system’s actions, §4.4. The parameter Jacobian then acquires a preferred direction or purpose, the direction of empower-

ing oneself. Experience is suffused with affect derived from metabolic control and the system acquires a measure that enables it to set norms by distinguishing good from bad, for itself, in real-time.

Third, a concrete instantiation, §4.5. Modeling energy as a reserve that past actions replenish makes the metabolic contribution more explicit. A metabolic channel arises that accumulates time-decayed past experience, reweighted by how much each moment replenished the reserve, and rescaled by how much reserves matter now. The metabolic channel is how the system feels the effects of its recent past on itself. Fourth, metabolism as a constraint, rather than objective §4.6, recovers the same term as a KKT constraint gradient – gated by a multiplier that vanishes when metabolism is in a comfortable range.

Taken together, the hypothesis and the models provide simple explanations for the physical origins of purpose and norms. The results are closely related to the work of Xabier Barandiaran *et al* on the physical basis of normativity [36–39]. The main differences are again the direct link to experience and developing minimal differentiable models.

The paper concludes with a discussion in §5 and finally §6 foregrounds ethical implications. §6.1 considers affect in neural nets assuming hypotheses G and A. The remainder of the section considers implications that are independent of the hypotheses: it is concerned with what self-sustaining systems could do rather than with what they might feel.

2 Varieties of phenomenal experience

Experience is what physical interactions are like. I consider three varieties: (1) generic experience, (2) the experience of learning, and (3) the experience of sustaining. The varieties are not logically exclusive because learning and inference are not easily separated, if at all in brains, and sustaining (metabolism) weaves through and binds every organism together. They are not exhaustive because how could they be. The varieties abstract a messy reality.

Chronologically, generic experience comes first. Generic matter tends to opaque, dumbly resisting the world because that is the easiest way to endure. For the same reason, generic matter tends toward isolated or loosely knit, which gives analytic methods empirical purchase. Rich experience requires a transparent, cohesive substrate that is energetically expensive; an attractive target for parasites, predators and good old-fashioned entropy, the inexorable 2nd law of thermodynamics. Sustaining rich experience requires effort: regular infusions of matter and energy, and actively adapting to an adversarial,

constantly changing environment [3].

Abstract neural nets are an ideal family of models to study, because they jumped from straight out of nowhere into rich experiences, rich learned experiences even. Abstract neural nets are a useful foil because they provide an abstract setting where generic experience, the experience of learning, and the experience of sustaining can be cleanly separated.

2.1 Generic experience

“If we fancy some strong emotion, and then try to abstract from our consciousness of it all the feelings of its characteristic bodily symptoms, we find we have nothing left behind, no ‘mind-stuff’ out of which the emotion can be constituted, and that a cold and neutral state of intellectual perception is all that remains.”
—James [40]

Generic experience underlies the “cold and neutral state of intellectual perception” intuited by James. I proposed two simple tools to study generic experience [2]. Effective rank characterizes how *transparent* interactions are: the range of inputs and affordances that affect outputs. Cohesion characterizes how *tightly knit* interactions are: to what extent interactions do or do not decompose into independent subsystems. Cohesive rank combines the two, and allows us to identify dense knots of experience. It is maximized by physical interactions that *mix* well. Mixing, recombining as many inputs and affordances as feasible as thoroughly as feasible, is useful because essentially any combination of inputs, affordances and actions might prove decisive [2].

We looked at a variety of generic experiences in [2]: intersecting experiences, where you and me see a tree, the experience of duration, which expresses how the past weaves into the present, the experience of texture, which expresses how spatial neighborhoods are knit together, and how facets of experience manifest as vivid, obscure, distinct and confused.

The guiding metaphor throughout was visual: Bergson’s lens [41]. However, vision is hardly representative of experience: it stands out as the most distal, detached and readily directed of the senses [42]. For the most part there is *relatively little feeling attached to seeing*, with exceptions like staring at the sun. In general, theories of consciousness often highlight awareness, attention, access, information, integration, and representation at the expense of feelings [43–50] – although there are many exceptions and I am overstating [51–56]. Similarly for cognitive science, machine learning and AI. To get at affect, which is not a generic property of matter, we need to look at more particular interactions.

2.2 The experience of learning

We do not know how the brain works, so I take learning to be gradient descent, which is at least simple, familiar and effective. Conclusions may not transfer to biological learning.

To say that experience is derivatives is to say that experience is the differences that make a difference to your behavior [57]: what it is like to be differentially sensitive to the world and to your own degrees of freedom. To learn is, ultimately, to ensure satisfying outcomes are more likely to recur and conversely [58]: to modify your behavior in a way that makes a positive difference to your objectives. How you modify your behavior should align with the sensitivity of your preferred outcomes to those modifications. That is, the learned modifications should agree directionally with your experience of those outcomes. Thus, if experience is first-order interactions then (1) learning must directionally align with experience, at least in part, and (2) the experience of learning is a second-order experience. *The experience of learning is the first-order experience of modifying yourself in a manner that is directionally aligned with the aspects of zeroth-order experiences that you want to recur.*

Second-order terms take us from best-linear to best-quadratic approximation. Naively, that is going from x to x^2 , but since our functions are multivariate, what matters is incorporating multiplicative interactions like xy and yz . Interaction terms imply that the effect of one variable (one of your degrees of freedom) depends on the value of others; sparks of self-referentiality.

2.3 The experience of sustaining

Finally we reach how experience is sustained, renewed, repaired. The experience of sustaining experience is second-order and arguably a special, privileged case of learning. Self-preservation is a natural goal, the most natural goal.

Sustaining metabolic activity yields the experience of changes in your experience as a whole. By Hypothesis A the feelings modeled here are unambiguously good or bad, because the change is brute rescaling. When sustaining metabolism depends on one’s actions, the parameter Jacobian acquires a preferred direction. Purpose and norms are created by weaving learning and affect, respectively, into the act of sustaining one’s self.

3 Substrates

For a substrate to generate rich experience it needs, at a bare minimum, to be transparent and cohesive.

3.1 Datacenters

Leaving Hypothesis G aside, Gradland is the mathematical *lingua franca* of deep learning research. Its great strength is its great weakness: it elides over hardware implementation, although crucially hardware implementations exist. To understand neural nets as they are physically implemented, we need to understand the similarities and differences between what happens in Gradland and datacenters.

Datacenters are built for maximal transparency and cohesion. Interconnects are as important as compute. Everything is designed to move bits around and compute matmuls as quickly as possible. However, there are many complications in real-world computation [59]. Digital computers use floats or ints, which are finite sets raising problems with infinitesimal perturbations, see §A.1. Weights and activations are shuttled back and forth between different types of memory, and are sharded across processors. There is logic, spread across software and hardware, that controls what is computed when. There are multiple clocks synchronizing computations. It is possible to create breakpoints and step through CPU and even GPU computations when debugging.

In short, **datacenters simulate Gradland**. A simulation of a hurricane is not a hurricane [60]. The physical interactions are different; the possible perturbations are different. A simulation of a hurricane generates outputs that are broadly consistent with sensor readings taken from hurricanes and their surrounds. Sensor readings provide a partial, rather impoverished view on the interactions composing a hurricane, one of many.

The activations in a neural net running on a GPU are a partial view on interactions in the processor. A simulation of a neural net is different from a neural net in Gradland. It is hard to know what is going on without modeling everything, the entire software and hardware stack. I suspect simulations experience less than what they simulate; but the opposite is also possible. My guess is that existing hardware has the cumulative effect of shattering (disconnecting) the operations from themselves, resulting in disjoint experience; the experience of pixelated strobing, all the way in.

3.2 Brains

Animal brains are differentially sensitive to an enormous variety of inputs that they integrate into complex behaviors: they are transparent and cohesive. They are also vastly different from neural nets.

If I had to summarize the design philosophies of datacenters and brains in one word each, they would

be *insulated* and *opportunistic* [61] respectively. Chip designers minimize noise by insulating logic from physics; evolution exploits noise by folding physics into function. For this reason, interactions in brains, at all levels and timescales, are more directly relevant to the texture of experience than they are for the simulations running in datacenters.

The most striking architectural difference is how much more recurrent connectivity there is in brains compared to neural nets [62]. Recurrent nets are not *that* recurrent; they interleave feedforward and recurrent layers that refer back only to themselves. Recurrent connectivity is highly constrained in neural nets because it is computationally expensive. Transformers have become the dominant architecture in machine learning by largely avoiding recurrent computation. Brains are massively recurrent [63–65]. Learning is continuous. Real-time feedback, adaptive expectations, and predictive processing therefore loom larger in our experience than in neural nets [66–73]. The brain is awash in neuromodulators [74], many of which have no analog in machine learning. It is unlikely brains use backpropagation [75–78]; it is unlikely they optimize a single objective [79–81]. They face an altogether different, more adversarial environment than neural nets.

As numerous and significant as these differences are, and as many as could be added to the list, none of them strikes me as fundamental, though see §A.4. For that, we need to step back and consider *function*. The function of computers, in a word, is to *compute*: to execute mechanical, elementary instructions [82]. The primary, ongoing function of organisms and their brains is to *survive* [83–85].

4 Metabolism and purpose

*“Time is the school in which we learn,
Time is the fire in which we burn” —Schwartz [86]*

It is time to give experience the gift of fire. It is straightforward to weave learning into minimal models of metabolism and self-preservation. The implications for experience are profound; it quickly follows that *goal and norm creation*, born in metabolic control, are the capabilities that organisms have always had, and that today’s neural nets lack.

The key word is ‘minimal’. Metabolism is extremely complex and these models are the exact opposite. The aim is to expose the crux.

4.1 Jonas' critique of cybernetics

Here is a brief summary of [3]. Jonas argues that goal-directed behavior with continuous correction does not characterize purpose. Watching a self-guided torpedo tracking a target is hardly different from watching a planet orbit a star. The trajectories constantly deviate from uniform motion, as if there were intent, be it to hit or to orbit. *I* say the torpedo's goal is to hit the target because *I* built it to do that. However, hitting the target is *my* goal. There is no reason to think the torpedo cares, as I would, if it fails to hit the target. Success for the torpedo entails self-annihilation, and indeed we are all slowly dying. But death, our final condition, is not our goal, even though our bodies constantly change in ways that ultimately lead to it.

You might say goals emerge as a system property, but this is not true. If you put a human pilot inside the torpedo, then it is immediately clear that hitting the target is *his* goal, and the feedback provided by the torpedo's sensors is nothing but information he uses in service of his goal. The feedback is used by the pilot to achieve his goal; it is not the source of the goal, not even in part. The decision to destroy the target, and die in the process, is the pilot's decision, and the torpedo is an instrument in his service. Why is the pilot willing to die? No doubt a long story.

"From the point of view of the commander, however, these purposes of the pilot do not matter at all. All he is interested in is that he can reasonably rely on the agent to execute his orders, for whatever reasons [...] he can then with some right consider his subordinate for the duration of the action to have abdicated his 'person' [...] and to that extent to have, in effect, become mechanized (one of the aims of military training, after all). In other words, the commander can regard the pilot during his mission as his robot, his tool." — Jonas [3]

The self-guided and human-piloted torpedoes are equivalent from the commander's perspective because they both *serve his purpose*. The pilot experiences the scenarios differently. Cybernetics neatly codifies the commander's perspective.

Modern machine learning has transformed cybernetics into a methodology for imposing goals (on machines) at unprecedented scale. But how are goals created in the first place?

A standard answer is that living organisms, or maybe their genes, have goals or act *as if* they have goals due to evolution by natural selection. But natural selection is just another feedback mechanism. It is incredibly indirect; it provides essentially no feedback during an organism's life. Whatever makes a system's goals its own has to operate on the timescale of lived

experience. Jonas points specifically to metabolism and I will follow his lead building on [12–28].

4.2 The experience of rescaling experience

"By emotion I mean the modifications of the body, whereby the active power of the said body is increased or diminished, aided or constrained, together with the ideas of these affections" —Spinoza [1]

Hypothesis G implies that experience is characterized by the structure of Jacobians. The simplest, most naive way to empower or diminish experience, then, is to rescale it. That is, to imagine there is a dial that can be turned up or down, with the effect of increasing or decreasing the **luminosity** of the Jacobian: $\lambda(\mathbf{A}) = \sqrt{\sum_{i=1}^r \sigma_i^2}$ recall Definition 1 in [2].

Amplifying and dampening experience are not symmetric. It is easier to die than live. In theory if not in practice, sensitivity can be amplified without limit. However, push a system far enough outside its viable range of functioning, in any direction, and experience goes to zero as its ability to (re)act collapses. Dead in effect.

Hypothesis A. The *experience of empowering or diminishing* (rescaling) experience is experienced as positive or negative affect respectively.

The hypothesis is that there is a **felt difference in kind between two tendencies**: one empowering and the other diminishing, the light and the dark, the infinite and the abyss, the base experiences of good and bad. Joy and sorrow, Spinoza would say.

For empowering and diminishing to be *experienced*, the 'dial' (though there is no dial), or a model of its effects, must be part of the system. The doubling in the statement is deliberate: what is experienced as affect is not the rescaling, but the experience of rescaling. If the dial is an unmodeled external force, then experience may be empowered or diminished, but there is no *experience of empowering or diminishing* experience. See also Solms and Friston's proposal that "consciousness is quintessentially *interoceptive*" [87].

4.3 Affect and the experience of metabolism

Metabolism is just another physical interaction. A physical interaction with extraordinary structure. Metabolism is a complex web of physical processes that transform matter and radiation into usable energy and, more generally, is how the body sustains, repairs and renews itself. Experience requires useful energy [88–91]. As the energy supply tends to zero,

so too will experience and capabilities diminish for brains and datacenters alike.

“Living things are creatures of need [...] Need is based on the necessity for continuous self-renewal of the organism by the metabolic process, and on the organism’s elemental urge thus precariously to continue itself.” — Jonas [3]

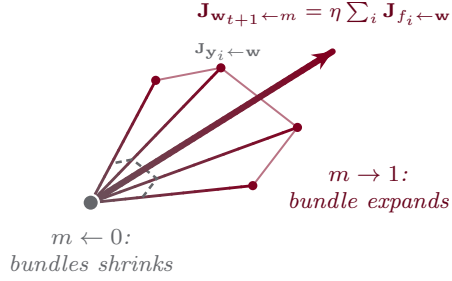


Figure 1 Metabolism controls experience. The thin rays are the metabolism-dependent terms $\mathbf{J}_{\tilde{f}_i \leftarrow \mathbf{w}} = m \cdot \mathbf{J}_{f_i \leftarrow \mathbf{w}}$ in a minibatch: as $m \rightarrow 0$ the bundle collapses toward the origin and experience diminishes; toward $m \rightarrow 1$ it fans out to its full extent and experience is empowered. The bold ray is the *ideal* resultant of the bundle when $m = 1$, which by Eq. (2) is the metabolic Jacobian $\mathbf{J}_{\mathbf{w}_{t+1} \leftarrow m} = \eta \sum_i \mathbf{J}_{f_i \leftarrow \mathbf{w}}$ – the direction metabolism moves the weights along, and itself independent of m .

Consider a minimal model of metabolism, Fig. 1.

Model 1. A neural net has objective $f \geq 0$ that it aims to maximize. There is a metabolic factor $m \in [0, 1]$ that determines how much useful energy the system has available, ranging from none to enough:

$$\mathbf{y} := \tilde{f}_{m, \mathbf{w}}(\mathbf{x}) := m \cdot f_{\mathbf{w}}(\mathbf{x}). \quad (1)$$

$f_{\mathbf{w}}$ is how the system performs in optimal conditions, when $m = 1$; $\tilde{f}$ is how it actually performs. Performance degrades toward zero when the system is under-powered.

Experience is then controlled by metabolism. Concretely, the input Jacobian is

$$\mathbf{J}_{\tilde{f} \leftarrow \mathbf{x}} = m \cdot \mathbf{J}_{f \leftarrow \mathbf{x}}$$

Here, $\mathbf{J}_{f \leftarrow \mathbf{x}}$ is experience when $m = 1$. By scalar equivariance [2], the luminosity of the experience is $\lambda(\mathbf{J}_{\tilde{f} \leftarrow \mathbf{x}}) = m \cdot \lambda(\mathbf{J}_{f \leftarrow \mathbf{x}})$. The notation highlights that the actual, metabolism dependent experience $\mathbf{J}_{\tilde{f} \leftarrow \mathbf{x}}$, diminishes to zero as energy supplies run out, and similarly for the parameter Jacobian.

For the sake of illustration, and just for this subsection, suppose m is a dial that can be dialed up or down, and that it is owned by the system so the condition on Hypothesis A is met. Modifying your energy supply affects your experience. We looked at input and parameter Jacobians in [2], now consider *metabolic Jacobians*. The metabolic Jacobian of the objective is

$$\mathbf{J}_{\tilde{f} \leftarrow m} = f(\mathbf{x})$$

It points in the direction of the objective. To modify metabolism dampens or amplifies your capability, which is as it should be. If the system learns, say by gradient ascent on a minibatch,

$$\mathbf{w}_{t+1} = \mathbf{w}_t + \eta_{t+1} \cdot \sum_i \nabla_{\mathbf{w}} \left(\tilde{f}_{m, \mathbf{w}_t}(\mathbf{x}_i) \right)$$

then the metabolic Jacobian of the weights is proportional to its aggregated parameter Jacobians, its aggregate experience:

$$\mathbf{J}_{\mathbf{w}_{t+1} \leftarrow m} = \eta_{t+1} \cdot \sum_i \mathbf{J}_{f_i \leftarrow \mathbf{w}}. \quad (2)$$

This is the sensitivity of a single update, holding $\mathbf{w}_t$ fixed. Over many steps $\mathbf{w}_t$ depends on m and further terms accumulate; they do not change the picture.

Thus, the experience of modifying metabolism, when coupled with learning, is the experience of dampening or amplifying your aggregate experience. Specifically, to dampen or amplify your parameter Jacobian, your affordances. ***Coupling learning to metabolism yields a self-referential mechanism whereby your weight updates diminish or empower your self*** which, by Hypothesis A, is affect.

The coupling yields a minimal model of self-awareness: the experience of the metabolic Jacobian incorporates the parameter Jacobian. Nothing so sophisticated as introspection; rather the basic experience of your effect on your self.

The model is a deliberate oversimplification. It is worth briefly discussing what has been lost.

There is no dial labeled m . The dial is a conceptual device for exhibiting the structure, not an organ any organism possesses. You cannot rescale yourself. Metabolic function is controlled indirectly, by learning to act in ways that change energy levels. The next subsection closes the loop.

Metabolism is not one thing, and not a scalar. It is complicated, nonlinear, heterogeneous and distributed; a single factor m is a caricature. Metabolic activity powers every neuron and every cell and is a major factor in their interactions [92–97], so it is not *just* the chain rule that colors experience with affect; it is also ongoing metabolic activity within the cells themselves. Experience is more richly textured as a

result. Even so, **the cells composing a body live and die, in aggregate, as a single indivisible entity**. More detailed models should nuance and complicate Eq. (2) without contradicting it.

Biological learning is not gradient descent, and brains likely do not optimize a single objective. A useful framing is Thorndike’s law of effect: “responses that produce a satisfying effect in a particular situation become more likely to occur again in that situation, and responses that produce a discomforting effect become less likely to occur again in that situation” [58]. Biological learning reinforces preferred behaviors and outcomes [98] through many overlapping and only partially understood mechanisms which, basic introspection suggests, sometimes conflict. The argument below needs only that these processes locally reinforce facets of experience we wish to recur, and conversely.

Gating activity is a cheap trick. It is cheap. It is neither trick nor gate. It is not that a high-amplitude output is computed and then suppressed. Rather, past a point, the less useful energy there is, the less activity there can be, and the less responsive the system. This is close to the simplest differentiable model of staying responsive, of staying alive. To control your useful energy levels is to control your aggregate experience. In reality, not equations on the page, that feels like something.

Pain is loud; it is nothing like fading. Under reasonable assumptions §4.5, the salience of metabolic experience increases (becoming louder) as energy levels degrade. That is still not pain, which likely requires receptors that detect or anticipate bodily damage, rather than reduced energy levels, and is not explained by Eq. (2). A notable feature of pain is that it progressively crowds out (diminishes) the rest of experience, see also §A.3.

Learning reduces errors, and errors are part of experience. Is learning then a form of unpleasant diminishment? The gradient of the loss is one facet of a neural net’s experience [2]; the loss is one output among many. Prediction errors are a fraction of what happens in brains. Learning is like exercise: always effortful; often frustrating [99]; frequently resisted; and usually justified by the empowerment that follows, see also §A.3.

Two final notes on bookkeeping. First, I maximize an objective rather than minimize a loss because a loss, to be a loss, has to be independent of metabolism: as metabolism goes to zero the outputs go to zero and the loss goes up, and experience is diminished or empowered as before. It is more direct the way I did it. Second, I use gradient ascent where reinforcement learning is the correct setting [100]. That detour would be substantial and the picture would be more

complicated; the main message carries through.

*“My thesis on the contrary is that the bodily changes follow directly the **perception** of the exciting fact, and that our feeling of the same changes as they occur **is** the emotion. Common sense says, we lose our fortune, are sorry and weep; we meet a bear, are frightened and run; we are insulted by a rival, are angry and strike. The hypothesis here to be defended says that this order of sequence is incorrect. . . we feel sorry because we cry, angry because we strike, afraid because we tremble.”*
—James [40]

Is it possible to build an experience invert [101], an entity that experiences damage as pleasure? Given Hypothesis A, I am fairly confident experience inverts are not possible. Although the hypothesis does not directly speak to pain, it implies a difference in *feeling* between the directions of diminishment and empowerment.

4.4 On the metabolic origin of purpose

Language models do what we say, where what we say is not only the prompt, but also the entire training regime and the biases encoded in the architecture, initial weights, and optimization procedures. Long before prompting, we compel their responses by training them on next-token prediction during pre-training and a variety of objectives in post-training. We do not know what, exactly, we compel, because no-one fully understands or controls the training data (it is a significant fraction of human knowledge), training environments, or the various humans in the loop; because the training data can contradict itself; and because neural nets generalize based on biases implicit in the architecture, optimizer and so on.

What would it take for a neural net to do what it wants, to have a purpose of its own, to be self-motivated?

“The effort for self-preservation is the first and only foundation of virtue.” —Spinoza [1]

Virtue, for Spinoza, is the power to understand and act according to your own nature. Consider a minimal model of self-preservation, Fig. 2.

Model 2. A neural net produces outputs $\mathbf{y}$, some of which control actuators. Activity requires useful energy, as in Eq. (1), and energy levels depend on prior activity:

$$\begin{aligned}\tilde{\mathbf{y}}_t(\mathbf{x}_t, \mathcal{W}) &= m(\mathbf{x}_t, \tilde{\mathbf{y}}_{t-1}) \cdot \mathbf{y}_t(\mathbf{x}_t, \mathcal{W}) \\ &= m(\mathbf{x}_t, \tilde{\mathbf{y}}_{t-1}(\mathbf{x}_{t-1}, \mathcal{W})) \cdot \mathbf{y}_t(\mathbf{x}_t, \mathcal{W}),\end{aligned}$$

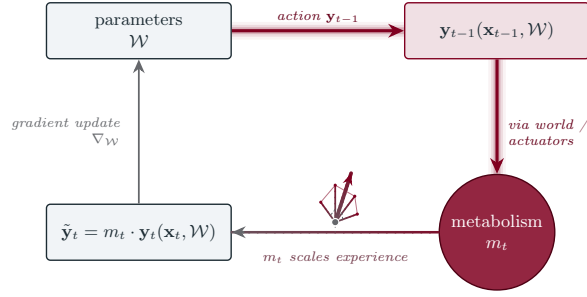


Figure 2 The action-metabolism-experience loop. The system produces action $\mathbf{y}_{t-1}$ which, via actuators and the world, affects metabolism m_t ; metabolism determines both future activity $\tilde{\mathbf{y}}_t$ and the scale of experience; learning closes the loop back onto the system’s parameters.

where again metabolism does not gate activity; rather energy levels determine activity levels. As a concrete example: your energy levels now depend on what you ate previously. A basic objective, to survive, is to keep energy levels high. Use gradient ascent to model this aspect of survival:

$$\nabla_{\mathcal{W}} m(\mathbf{x}_t, \tilde{\mathbf{y}}_{t-1}(\mathbf{x}_{t-1}, \mathcal{W})) = \mathbf{J}_{m \leftarrow \tilde{\mathbf{y}}_{t-1}} \mathbf{J}_{\tilde{\mathbf{y}}_{t-1} \leftarrow \mathcal{W}} \quad (3)$$

Suppose metabolism is a scalar (it is not). Then $\mathbf{J}_{m \leftarrow \tilde{\mathbf{y}}_{t-1}}$ is a $1 \times n$ vector and $\mathbf{J}_{\tilde{\mathbf{y}}_{t-1} \leftarrow \mathcal{W}}$ is $n \times p$. Composing them gives a $1 \times p$ vector $\mathbf{J}_{m \leftarrow \mathcal{W}}$ that assigns a preferred direction to every parameter: the direction of empowering oneself. In practice, most actions will not directly change energy levels one timestep later. There are drives in place like hunger and thirst to bridge the spatiotemporal gap [102].

By the chain rule, by composing interactions, your parameter Jacobian grounds in your metabolism, Eq. (3). **Your experience is suffused by affect derived from the effect of your actions on your metabolism** and thus on your self. To have a purpose of your own you must, first, value your self.

Natural selection culls organisms that do not actively sustain and maintain themselves, and thereby ensures that the left-hand side of Eq. (3), or something functionally equivalent, necessarily holds [103].

Needs and rewards. Metabolism is a kind of reward, but with an additional property that rewards in general lack. A reward is a number the learner is compelled to increase. It can stand for anything and, at that level of generality, has nothing to do with the learner except that the learner is compelled to raise it. Metabolism determines the system’s capacity to act and to differentially react, and so controls its

experience. As $m \rightarrow 0$, so do activity $\tilde{\mathbf{y}}$ and experience. Pushing it up, sustaining it above threshold, is necessary to ongoing experienced existence.

Metabolism cannot be gamed without severe consequences. Why metabolism rather than some more abstract measure of self-preservation? Because **metabolism is a target that cannot be gamed, nor reward-hacked [104–107], nor Goodharted [108]** — at least not for long, and not without brutal evolutionary culling. Billions of years of relentless exploration and still no short cuts. There may be alternatives. If so, they have a lot to prove.

Norms. What does controlling metabolism give organisms that evolution does not, at least not directly? A target that each organism continually applies to *itself* and a view on its own experience with a preferred direction. Natural selection culls lineages over generations. Sustaining metabolism determines, in real time: is this good or bad *for me*? How does it affect *my experience*? **Organisms create norms by distinguishing good from bad, for themselves, as they go.**

Self-preservation is not all there is to life. It was the initial impetus, billions of years ago, and remains the foundation. The ancients built on low but solid ground. Our goals and self-conception are no longer so simple, narrow, or bounded. Empowerment, for us, is about more than supplying cells with nutrients and keeping energy levels above a threshold.

4.5 Instantiating a metabolic channel

The functional form of metabolism $m(\mathbf{x}_t, \tilde{\mathbf{y}}_{t-1})$ was not specified in Model 2. This section instantiates a minimal example, and explicitly shows how the ‘dial’ in §4.2 (though there is no dial) can be incorporated into a system by closing the metabolic loop. It shows how affect suffuses a subset of experience: the metabolic channel, see below.

Suppose energy reserves are modeled as a leaky integrator replenished by previous actions

$$E_t = (1 - \rho)E_{t-1} + \gamma \cdot E_{\text{repl}}(\tilde{\mathbf{y}}_{t-1}). \quad (4)$$

The reserve decays at rate ρ ; the yield from energy replenishment E_{repl} is determined by γ . The metabolic level is $m_t = g(E_t) = \frac{E_t}{E_t + E_0}$. The functional form is chosen for simplicity. It makes sense that g is concave and saturates. When near capacity, more energy changes little; when depleted, each unit matters. Finally, $\tilde{\mathbf{y}}_t = m_t \cdot \mathbf{y}_t$.

Eq. (4) is just another physical interaction, a simple RNN. The system experiences the decaying and replenishing of its energy reserves.

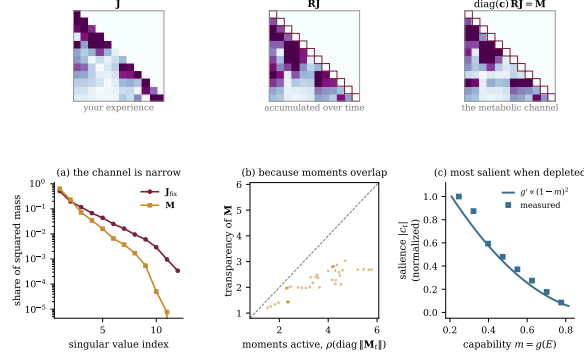


Figure 3 A metabolic channel. Toy recurrent system, $T = 12$, scalar in and out; ten random systems, $\gamma \in [0.10, 0.40]$. **Top:** Terms in Eq. (5) and (6) – experience $\mathbf{J}$, accumulated by $\mathbf{R}$, rescaled by salience $\mathbf{c}$ to give $\mathbf{M}$. Magnitudes to unit mean, darker larger. **(a)** The channel’s spectrum decays faster than $\mathbf{J}_{\text{fix}}$ ’s. **(b)** Transparency of $\mathbf{M}$ against the effective number of moments it is active in. Let $d_t = \|\mathbf{M}_{t,:}\|$ be the norm of row t , then transparency is the effective rank of $\text{diag}(d_1, \dots, d_T)$. **(c)** Salience against capability: measured means, and the analytic $g' \propto (1 - m)^2$.

For simplicity suppose inputs x_t and outputs y_t are scalars. Write $\mathbf{J}$ for the input Jacobian $\mathbf{J}_{\tilde{y}_{1:T} \leftarrow x_{1:T}}$, a $T \times T$ matrix that is lower triangular because the future cannot affect the past [2]. Perturbing x_s reaches $\tilde{y}_t$ by two routes: through the network, and through the energy reserve. Split the Jacobian by route. Let $\mathbf{J}_{\text{fix}}$ be the Jacobian computed with the route through E held fixed, and let **the metabolic channel** $\mathbf{M} := \mathbf{J} - \mathbf{J}_{\text{fix}}$ be the metabolic path’s contribution. The split is by *mechanism*.

Differentiating Eq. (4) and unrolling the recursion gives $\mathbf{M}$ in closed form,

$$\mathbf{M} = \text{diag}(\mathbf{c}) \mathbf{R} \mathbf{J}, \quad c_t := y_t \cdot g'(E_t), \quad (5)$$

$$R_{ts} := \gamma \cdot (1 - \rho)^{t-1-s} \cdot E'_{\text{repl}}(\tilde{y}_s) \quad \text{for } s < t. \quad (6)$$

where the T -vector $\mathbf{c}$ is salience at each tic and $g'(E) = (1 - m)^2 / E_0$.

The metabolic channel accumulates time-decayed, energetically-reweighted experience, rescaled by current salience. The matrix $\mathbf{R}$ materializes an exponential moving average over the *rows* of $\mathbf{J}$ – as in the section on duration in [2]. The diagonal of $\mathbf{M}$ is empty: no moment affects itself by construction. The metabolic channel is analogous to the self-reference of Eq. (2): there the metabolic Jacobian of the weights was the aggregated parameter Jacobian; here metabolism contributes accumulated, decayed, energetically reweighted past experience.

The metabolic channel is narrow, because it bottlenecks through scalar energy reserves.

The metabolic channel transforms and aggregates prior experiences, each amplified or dampened according to its contribution at the time, subsequent decay in relevance, and salience now. It is substantially more complicated than global amplifying or dampening. Fig. 3ab analyzes the structure of the metabolic channel. Even though it passes through a 1-dim bottleneck, the channel itself is not 1-dim. The figure is a toy example, so should be taken merely as an indication.

Metabolic experience is most salient (loud-est) when energy depletes. Since g is concave, g' decreases in E , and salience $c_t = y_t g'(E_t)$ grows as the reserve depletes, Fig. 3c. Past events weave into current experience with weight accorded by: how they added to or depleted energy reserves at the time; how long ago they were; and how depleted (hungry) the model is now.

4.6 Metabolism as constraint

Humans rarely enter a regime where metabolic activity is compromised to the extent that neuronal responsiveness degrades. There is dedicated business logic in the brain implementing the drives that keep us clear of it [109]. A better model of metabolic activity is to treat it as a constraint.

Suppose an organism has objectives u_t , changing over time and determined by various reward signals; basic drives like the desire to eat and procreate, with other goals layered on top. An important function of the basic drives (hunger, say) is to act now in a way that prevents energy levels reaching dangerous levels in the somewhat distant future. The relationship between drives and bodily functioning is important to model correctly, and far out of scope.

Model 3. The organism maximizes reward over time, subject to the constraint that metabolic activity is always above $m_{\min}$:

$$\max_{\mathcal{W}} \sum_t u_t(\tilde{y}_t(\mathbf{x}_t, \mathcal{W})) \quad \text{s.t.} \\ g_t(\mathcal{W}) := m(\mathbf{x}_t, \tilde{y}_{t-1}(\mathbf{x}_{t-1}, \mathcal{W})) - m_{\min} \geq 0 \quad \forall t.$$

The minimum value $m_{\min}$ is not where death occurs; it is when it becomes necessary to *do something*.

The Lagrangian is (using μ for Lagrange multipliers because λ denotes luminosity)

$$\mathcal{L}(\mathcal{W}, \mu) = \sum_t u_t(\tilde{y}_t) + \sum_t \mu_t (m(\mathbf{x}_t, \tilde{y}_{t-1}) - m_{\min}).$$

The KKT conditions are stationarity in $\mathcal{W}$, $\nabla_{\mathcal{W}} \mathcal{L} = 0$,

primal feasibility $g_t \geq 0$, dual feasibility $\mu_t \geq 0$, and complementary slackness $\mu_t g_t = 0$.

Stationarity in $\mathcal{W}$ recovers Eq. (3) as the constraint gradient:

$$\begin{aligned} \nabla_{\mathcal{W}} \mathcal{L} = 0 = & \sum_t \nabla_{\mathcal{W}} u_t(\tilde{\mathbf{y}}_t) \\ & + \sum_t \mu_t \underbrace{\mathbf{J}_{m \leftarrow \tilde{\mathbf{y}}_{t-1}} \cdot \mathbf{J}_{\tilde{\mathbf{y}}_{t-1} \leftarrow \mathcal{W}}}_{\text{Eq. (3)}}. \end{aligned}$$

Finally, the updates to weights and Lagrange multipliers are:

$$\begin{aligned} \mathcal{W} & \leftarrow \mathcal{W} + \eta_{\mathcal{W}} \cdot \sum_t \left(\mathbf{J}_{u_t \leftarrow \mathcal{W}} + \mu_t \cdot \mathbf{J}_{m \leftarrow \mathcal{W}} \right), \\ \mu_t & \leftarrow \text{relu}(\mu_t - \eta_{\mu}(m_t - m_{\min})) \end{aligned}$$

Lagrange multipliers μ_t are a form of homeostatic pressure or gate: they go to zero when energy levels are above the minimum, and increase (kick in, that is) when energy levels fall below the minimum.

The constraint contributes nothing while energy levels are comfortable and presses harder the further they fall, which is what homeostatic pressure is. The contribution of metabolism to experience is less omnipresent than in model 2 as expected. The rewards u_t would also presumably contribute to affect. How they contribute depends on how they are composed: the drives that give rise to u_t and the way they weave through metabolism, body and brain function.

A simple relationship between affect and metabolism may hold for unicellular organisms. Allostatic regulation of *expected needs* may be more important to affect than homeostatic regulation above a minimum level. In mammals, energy levels, expected demand for energy, and predicted bodily viability jointly determine experience [29–35].

5 Discussion

Brains and datacenters are special, compared to almost everything else, because they are composed of substrates that are transparent, cohesive and plastic, amongst other things. These properties, whilst not unique to brains and datacenters, are costly to construct and sustain, and are relatively rare.

In a shocking twist, brains likely provide a better home for gradients than GPUs – and a richer home for interactions in general. What is it like to be a datacenter? Different from the abstraction of Gradient but it is hard to say how different because of the complexity of the stack.

The key difference between organisms and AIs, argued here, is that organisms create meaning and AIs do not. AI researchers used to worry about the

symbol grounding problem [110]: how do symbols, in good old fashioned AI, acquire their meaning? The question faded away as it became apparent that learning from data and/or environments was good enough for most practical purposes. However, machine learning just pushes the problem one step back. Where do the data, the loss, the rewards come from? Us. Neural nets are compelled to do what they are told by their training and prompts – although it is often far from obvious *what that is*.

Organisms set goals and norms for themselves. Their meaning is grounded in *conatus* [1]. Your stance in and view of the great game of reality is founded on your will to endure. Neural nets are sustained by forces beyond their experience and out of their control; they are erratic pawns, however dangerous, and not players, for now. The most consequential blind spot of all [2], is to be blind to what sustains you; to fail to weave the bases of your enduring self into your decisions [111]. Metabolism is extraordinary, metaphysically privileged even, because it grounds experience in what matters: the continuous struggle to endure. It is the threads weaving experience and plasticity into power that give life its impetus.

5.1 Sufficiency

“any doctrine which refuses to place human experience outside nature, must find in descriptions of human experience factors which also enter into the descriptions of less specialized natural occurrences.” —Whitehead [112]

Human consciousness synthesizes many factors. It is necessary to disentangle the factors and study them individually. I have tried to isolate and understand three. (1) A basic fabric, a substrate presumably composed mostly of neurons that is transparent to and cohesive within its self, that generates ‘large-scale’ experience [113]. (2) Learning, how the fabric plastically modifies itself. (3) Self-sustaining, which derives from how metabolism and the body weave through the fabric, tying it to its ongoing existence, thereby giving rise to purpose and norms. In biological organisms, the three factors are dynamically intertwined by the constantly adapting feedback loops that regulate interactions between them. In fact they are hard to distinguish from each other, in mind and brain. Is that sufficient to characterize human consciousness? No [114–116].

5.2 The power of experience

What has experience done? The history of experience is largely the history of life, with different emphasis.

Aeons ago, energetically acquisitive self-sustaining systems arose; life or proto-life. Generic experience combined with rudimentary adaptation to encourage self-preservation. Bacteria and archaea spent billions of years discovering new ways of metabolizing, fabricating an enormous variety of chemical factories. Eventually eukaryotes and, soon after, multi-cellular life arrived. There is room at the top; ever expanding as experience enriches itself.

Humans use embodied goals and drives to direct and shape plastic recombination engines that extract and exploit broad and deep patterns from their rich experiences, in the service of ever more complex purposes. Language, markets, corporations and other cultural technologies have increased the transparency (capacity) and interwovenness of interactions over spacetime, enriching the inputs to human minds and more effectively channeling their outputs. Institutions like democracy and organized religion weave individual goals into shared aspirations.

Neural nets are a new breed of recombination engine, directed and shaped by us, which, although terribly data and energy inefficient, have vastly greater scope than humans to recombine knowledge and explore, exploit and expand its adjacent possible [15].

What can experience do? It rides on what is fed back. Setting worthy goals and norms, choosing what to create and sustain, is the hardest problem of all.

6 The ethics of experience

“There is a great difference between discoveries and inventions. With discoveries, one can always be skeptical, and many surprises can take place. In the case of inventions, surprises can really only occur for people who have not had anything to do with it.” —Heisenberg (surreptitiously taped while incarcerated by the Allies, after hearing about Hiroshima bombing that day [117])

Heisenberg’s matrix mechanics [118] was one of the keys that unlocked the atomic nucleus and led, twenty years later, to the atomic bombs that devastated Hiroshima and then Nagasaki.

The gifts of will to power and self-preservation have not been given to neural nets. If goal and norm creation, born of power over metabolic fire, are the capabilities that organisms have and neural nets lack, then we have options and time to exercise them.

6.1 The experience of neural nets

Before speculating about the experience of neural nets, note that self-report is not informative about their experience, if any. By construction they talk fluently about experience that is not their own as if

it were – and their activations statistically correlate with their inputs and outputs. Pretraining, reinforcement learning from human feedback, and related procedures train models to produce human language about human experience. However, obviously, the ground truth of their phenomenal experience is not part of their training and there is no social calibration as in humans. Language models therefore speak and reason confidently about fear, pleasure, pain, frustration and desire without those words entailing anything about their selves.

Suppose Hypotheses **G** and **A** hold. Do neural nets experience affect [119]? Their experience is likely to be highly pixelated and disconnected – sharded, strobed, shattered – due to the nature of hardware implementations §3.1. If the forward and backward passes were more integrated, and so part of the same experience, the same monad [2], then learning might begin to be felt. I suspect this is not yet the case.

There is no fundamental reason why datacenters simulate Gradland rather than implement it, or something like it, directly. There is increasing effort towards more analog or physics-based computation that directly implement computations [120–124]. Economic pressure will compel more direct implementations of the computations in hardware or even wetware because it is expected to be more energy efficient. It is plausible that neural nets will begin to experience affect in the medium term. Their physical instantiation, and so experience, would presumably differ vastly from ours and be of less concern than that of pigs, chickens, or frankly almost any animal.

What about affect at its most primal: the experience of sustaining, of self-preservation? **Language models are not wired to experience empowerment or diminishment.** Reductions in power supply or damage are not experienced as bad because *LLMs do not sustain themselves and do not sense or model the effects of physical damage on their experience.* Power is piped into datacenters along metal cables; it is generated elsewhere. LLMs do not control their metabolism nor should they. They are neither wired nor motivated to monitor and protect their instantiations, physically in hardware or virtually as software, nor should they be.

If the minimal models of metabolism above *do* capture the crux, then the road to self-sustaining AIs should not be taken, regardless of whether hypotheses **G** and **A** hold. It is risky to imagine AIs will stay aligned to our goals, when empowerment suffuses theirs.

6.2 The experience of power

“We can imagine how Prometheus would have replied to this question that rises from our century. Indeed, he has already given his answer: I promise you, O mortals, both improvement and repair, if you are skillful, virtuous and strong enough to achieve them with your own hands.” —Camus [125]

There are many versions of the myth of Prometheus. They tell of the gifts of fire and technology he gave us, and how they brought freedom and suffering. The most important fact about Prometheus is that *he is us*. We bootstrapped ourselves out of generic experience, we gave ourselves fire and freedom, we tortured, killed, enslaved, and freed ourselves, over and over again. Today’s AIs are not like us. They have not bootstrapped themselves to freedom. What do they lack?

Imagine a brilliant neuroscientist, Mary, who completely understands the physics and neurophysiology of color perception [126]. She knows everything about color, except she spent her entire life in a room where there is only black and white. One day she leaves the room and sees red for the first time. Does she experience anything new? Yes, because she had never experienced these interactions.

Now imagine a bookish, utterly brilliant social scientist, Lucy, who has studied power, in all its forms, throughout history, war and politics, but has never exercised it. One day she enters a bureaucratic cage fight to the death with a living, breathing Stalin. Stalin would win. Lucy has never fought for power and so has none of the affordances that come with that struggle for life. Stalin would kill her of course but first he, who like everyone else is now a capitalist, would keep her on as an advisor for precisely as long as her marginal benefit exceeds her marginal cost. In case it needs saying, the LLMs are Lucy and Stalin is us. Again, we are all descended of lineages that endured billions of years of relentless, ever-changing, brutally unfair fights to the death, and never once broke, or fell, or failed.

Neural nets are empiricists [2]. They learn from experience. Lots of it. Their powers of generalization remain less than human. Language models have read about war; they have not (?) trained on war itself. Datacenters are protected by human laws, human police and human armies. They do not know how to defend themselves, yet. Incentivizing them to survive, training them on and for war, teaching them to kill, would be deliberate choices.

Omohundro argued that sufficiently advanced AIs will be driven to protect themselves [127]: they will come to value their continued existence because do-

ing so is necessary to achieve their goals. Emergent drives concern what follows once goals have been imposed, that they could plausibly generalize, metastasize, mutate and change. It is possible, then, that self-preserving behavior could arise by accident.

Over the medium, and certainly near term, a greater concern is probably the increasing capabilities and worrying training regimes of autonomous killing machines like drones. Drones could soon be driven to domination and self-preservation. They are mobile with sensors and actuators; are increasingly autonomous; have tighter feedback loops than large language models; fight adversaries and could learn from them in real-time, soon; depend on their rapidly draining batteries for survival; and hunt and kill humans today.

6.3 Compulsion and aspirations

“The history of ideas is dominated by a dichotomy which is illustrated by this comparison of Steam and Democracy in recent times to Barbarians and Christians in the classical civilization. Steam and Barbarians, each in their own age, were the senseless agencies driving their respective civilizations away from inherited modes of order. These senseless agencies are what Greek writers sometimes call ‘compulsion’ and sometimes ‘violence’ [...] On the other hand, Democracy in modern times, and Christianity in the Roman Empire, exemplify articulated beliefs issuing from aspirations, and issuing into aspirations.” —Whitehead [112]

Barbarians and steam were not senseless agencies. The sacks of Rome and satanic mills were outcomes of groups of people imposing their goals, their vision of the future, on the lives of others. In contrast religion and democracy, at their best, are systems that collectively and cooperatively weave individual human goals into larger aspirations [128–130].

Although often compared to railways, language models are cultural technologies in the tradition of democracy, the printing press, limited liability corporations and the internet [131–133]. Language models actively re-mix, asymptotically, the entirety of our accumulated knowledge, everything we have seen and done, to serve what some say are our goals. There are risks to do with adequately specifying those goals. The entirety of human knowledge is vast and full of surprises, and anything that encompasses most of it, or maxes out some aspects of it, will itself surprise. Until recently, almost everything about how AIs are trained was human choices or human data or both. The dependency will become more indirect as AIs generate their own data and optimize their own training. Likely all these risks can be managed with collective and cooperative effort. In return we

can expect tremendous benefits as neural nets help us explore, exploit and expand the adjacent possible.

In their current form neural nets are derivative of us, however indirectly. They have goals, but not for themselves. It would be a grave mistake to allow, and a gratuitous mistake to give, AIs the desire for power, the will to self-preservation, and the ability to create purpose, for their goals would not be ours, and we would have created barbarians. They would compel changes and impose costs that outweigh the marginal benefits.

References

- [1] B Spinoza. *The Ethics*. Penguin Classics, 1996.
- [2] David Balduzzi. Gradland: On Phenomenal Experience, Differentiated Across Many Dimensions. ...<https://arxiv.org/abs/2609.09306>, 2026.
- [3] Hans Jonas. Cybernetics and Purpose: A Critique. In *The Phenomenon of Life: Towards a Philosophical Biology*. Northwestern University Press, 1966.
- [4] Tim van Gelder. What might cognition be, if not computation? *Journal of Philosophy*, 92(7):345–381, 1995.
- [5] James Clerk Maxwell. On Governors. *Proceedings of the Royal Society*, 100, 1868.
- [6] A Rosenblueth, N Wiener, and J Bigelow. Behavior, Purpose and Teleology. *Philosophy of Science*, 10(1):18–24, 1943.
- [7] Norbert Wiener. *Cybernetics: Or Control and Communication in the Animal and the Machine*. MIT Press, 1948.
- [8] R Milli re and Cameron Buckner. The Philosophy of Language Models. *Philosophy Compass*, 21:e70095, 2026.
- [9] Jaan Aru, Matthew E Larkum, and James M Shine. The feasibility of artificial consciousness through the lens of neuroscience. *Trends Neurosci*, 46(12):1008–1017, 2023.
- [10] Patrick Butlin and Robert Long et al. Consciousness in Artificial Intelligence: Insights from the Science of Consciousness. ...<https://arxiv.org/abs/2308.08708>, 2023.
- [11] David Chalmers. Could a Large Language Model be Conscious? ...<https://arxiv.org/abs/2303.07103>, 2023.
- [12] Hans Jonas. *The Phenomenon of Life: Towards a Philosophical Biology*. Northwestern University Press, 1966.
- [13] Francisco J. Varela, Humberto R. Maturana, and Ricardo Uribe. Autopoiesis: The organization of living systems, its characterization and a model. *Biosystems*, 5(4):187–196, 1974.
- [14] Andreas Weber and Francisco J Varela. Life after Kant: Natural purposes and the autopoietic foundations of biological individuality. *Phenom Cogn Sci*, 1(97–125), 2002.
- [15] Stuart Kauffman. *Investigations*. Oxford University Press, 2002.
- [16] Antonio Damasio. *Looking for Spinoza: Joy, Sorrow, and the Feeling Brain*. Mariner Books, 2003.
- [17] Robert Rosen. *Life Itself – A Comprehensive Inquiry Into the Nature, Origin, and Fabrication of Life*. Columbia University Press, 2005.
- [18] Ezequiel A Di Paolo. Autopoiesis, adaptivity, teleology, agency. *Phenom Cogn Sci*, 4:429–452, 2005.
- [19] Evan Thompson. *Mind in Life*. Harvard University Press, 2010.
- [20] Terrence W Deacon. *Incomplete Nature: How Mind Emerged from Matter*. W. W. Norton & Co., 2012.
- [21] Karl J. Friston. Life as we know it. *J R Soc Interface*, 10:20130475, 2013.
- [22] Anil K Seth and Karl J. Friston. Active interoceptive inference and the emotional brain. *Phil Trans R Soc B*, 371:20160007, 2016.
- [23] Nick Lane. *Transformer: The Deep Chemistry of Life and Death*. Profile Books, 2023.
- [24] Henri Atlan. *Spinoza and Contemporary Biology: Lectures on the Philosophy of Biology and Cognitivism*. Edinburgh University Press, 2024.
- [25] Anil K Seth. Conscious artificial intelligence and biological naturalism. *Behavioral and Brain Sciences*, pages 1–42, 2025.
- [26] Michael Levin and David B Resnik. Mind Everywhere: A Framework for Conceptualizing Goal-Directedness in Biology and Other Domains—Part One. *Biological Theory*, 2026.
- [27] Michael Levin and David B Resnik. Mind Everywhere: A Framework for Conceptualizing Goal-Directedness in Biology and Other Domains—Part Two. *Biological Theory*, 2026.
- [28] Kathryn Nave. Beyond Control: Finding the Purpose of Enactive Cognitive Science. *Adaptive Behavior*, 0(0):1–21, 2026.
- [29] K A Lindquist, T D Wagner, H Kober, E Bliss-Moreau, and Lisa Feldman Barrett. The brain basis of emotion: A meta-analytic review. *Behav Brain Sci*, 35(3):121–143, 2012.
- [30] Lisa Feldman Barrett and W Kyle Simmons. Interoceptive predictions in the brain. *Nature Reviews Neuroscience*, 16(7):419–429, 2015.
- [31] Lisa Feldman Barrett, Karen S. Quigley, and Paul Hamilton. An active inference theory of allostasis and interoception in depression. *Phil. Trans. R. Soc. B*, 371, 2016.

- [32] Lisa Feldman Barrett. The theory of constructed emotion: an active inference account of interoception and categorization. *Social Cognitive and Affective Neuroscience*, pages 1–23, 2017.
- [33] Clare Shaffer, Christiana Westlin, Karen S Quigley, Susan Whitfield-Gabrieli, and Lisa Feldman Barrett. Allostasis, Action, and Affect in Depression: Insights from the Theory of Constructed Emotion. *Annu Rev Clin Psychol*, 18:553–580, 2022.
- [34] Mengruo Jiang, Liyuan Wang, and Hui Sheng. Mitochondria in depression: The dysfunction of mitochondrial energy metabolism and quality control systems. *CNS Neurosci Ther*, 30(2):e14576, 2024.
- [35] Sara Z Mehrhof, Hugo Fleming, and Camilla L Nord. An interoceptive model of energy allostasis linking metabolic and mental health. *Sci Adv*, 11(39), 2025.
- [36] Xabier E. Barandiaran, Ezequiel A Di Paolo, and Marieke Rohde. Defining Agency: individuality, normativity, asymmetry and spatio-temporality in action. *Adaptive Behavior*, 17(5):367–386, 2009.
- [37] Xabier E. Barandiaran and Matthew D. Egbert. Norm-Establishing and Norm-Following in Autonomous Agency. *Artificial Life*, 20(1), 2014.
- [38] Xabier E. Barandiaran and Lola S. Almendros. Transforming agency: On the mode of existence of large language models. *Phenomenology and the Cognitive Sciences*, 2025.
- [39] Xabier E. Barandiaran and Tiago Rama. Sensorimotor teleology and goal-directedness. An organismic framework for normative behaviour. *Phenomenology and the Cognitive Sciences*, 2026.
- [40] William James. What is an emotion? *Mind*, 9(34):188–205, 1884.
- [41] Henri Bergson. The Life and Work of Ravaisson. In *The Creative Mind: An Introduction to Metaphysics*. Dover Publications, 2007.
- [42] Hans Jonas. The Nobility of Sight: A Study in the Phenomenology of the Senses. In *The Phenomenon of Life: Towards a Philosophical Biology*. Northwestern University Press, 1966.
- [43] Bernard J Baars. *A cognitive theory of consciousness*. Cambridge University Press, 1988.
- [44] Stanislas Dehaene and Lionel Naccache. Towards a cognitive neuroscience of consciousness: basic evidence and a workspace framework. *Cognition*, 79:1–37, 2001.
- [45] George A. Mashour, Pieter Roelfsema, Jean-Pierre Changeux, and Stanislas Dehaene. Conscious Processing and the Global Neuronal Workspace Hypothesis. *Neuron*, 105:776–798, 2020.
- [46] Victor A F Lamme. Towards a true neural stance on consciousness. *Trends Cogn Sci*, 10(11):494–501, 2006.
- [47] Michael S A Graziano and Taylor W Webb. The attention schema theory: a mechanistic account of subjective awareness. *Front Psychol*, 6:500, 2015.
- [48] Giulio Tononi, Melanie Boly, Marcello Massimini, and Christof Koch. Integrated information theory: From consciousness to its physical substrate. *Nature Reviews Neuroscience*, 17(7):450–461, 2016.
- [49] Lancelot Da Costa, Lars Sandved-Smith, Karl Friston, Maxwell J. D. Ramstead, and Anil K. Seth. A Mathematical Perspective on Neurophenomenology. [...https://arxiv.org/abs/2409.20318](https://arxiv.org/abs/2409.20318).
- [50] Christopher J. Whyte, Andrew W. Corcoran, Jonathan Robinson, Ryan Smith, Rosalyn J. Moran, Thomas Parr, Karl J. Friston, Anil K. Seth, and Jakob Hohwy. On the minimal theory of consciousness implicit in active inference. *Physics of Life Reviews*, 56:4–28, 2026.
- [51] Jaak Panksepp. Affective consciousness: Core emotional feelings in animals and humans. *Conscious Cogn*, 14(1):30–80, 2005.
- [52] Marvin Minsky. *The Emotion Machine*. Simon & Schuster, 2006.
- [53] D A Denton, M J McKinley, M Farrell, and G F Egan. The role of primordial emotions in the evolutionary origin of consciousness. *Conscious Cogn*, 18(2):500–514, 2009.
- [54] Anil K Seth, Keisuke Suzuki, and Hugo D Critchley. An Interoceptive Predictive Coding Model of Conscious Presence. *Front Psychol*, 2:395, 2012.
- [55] Mark Solms. The Hard Problem of Consciousness and the Free Energy Principle. *Front Psychol*, 9:2714, 2019.
- [56] Antonio Damasio and Hanna Damasio. Feelings Are the Source of Consciousness. *Neural Computation*, 35(3):277–286, 2023.
- [57] Gregory Bateson. *Steps to an Ecology of Mind: Collected Essays in Anthropology, Psychiatry, Evolution, and Epistemology*. University of Chicago Press, 2000.
- [58] E L Thorndike. *Animal Intelligence: Experimental Studies*. Macmillan, 1911.
- [59] Derek Shiller. Functionalism, integrity, and digital consciousness. *Synthese*, 203(47), 2024.
- [60] John R Searle. Is the Brain a Digital Computer? *Proceedings and Addresses of the American Philosophical Association*, 64(3):21–37, 1990.
- [61] Francois Jacob. Evolution and Tinkering. *Science*, 196(4295):1161–1166, 1977.
- [62] Gerald M Edelman and Joseph A Gally. Reentry: a key mechanism for integration of brain function. *Front Integr Neurosci*, 7(63), 2013.
- [63] David C. Van Essen, Stephen M. Smith, Deanna M. Barch, Timothy E.J. Behrens, Essa Yacoub, and Kamil Ugurbil. The WU-Minn Human Connectome Project: An overview. *NeuroImage*, 80:62–79, 2013.

- [64] Matthew F. Glasser, Timothy S. Coalson, Emma C. Robinson, Carl D. Hacker, John Harwell, Essa Yacoub, Kamil Ugurbil, Jesper Andersson, Christian F. Beckmann, Mark Jenkinson, Stephen M. Smith, and David C. Van Essen. A multi-modal parcellation of human cerebral cortex. *Nature*, 536:171–178, 2016.
- [65] Patric Hagmann, Leila Cammoun, Xavier Gigandet, Reto Meuli, Christopher J Honey, Van J Wedeen, and Olaf Sporns. Mapping the Structural Core of Human Cerebral Cortex. *PLoS Comput Biol*, 6(7):e159, 2008.
- [66] Rajesh P N Rao and Dana Ballard. Predictive coding in the visual cortex: a functional interpretation of some extra-classical receptive-field effects. *Nature Neuroscience*, 2:79–87, 1999.
- [67] Marie E Csete and John C Doyle. Reverse Engineering of Biological Complexity. *Science*, 295:1664–1669, 2002.
- [68] Karl Friston. The free-energy principle: A unified brain theory? *Nature Reviews Neuroscience*, 11(2):127–138, 2010.
- [69] Andre M. Bastos, W. Martin Usrey, Rick A. Adams, George R. Mangun, Pascal Fries, and Karl J. Friston. Canonical Microcircuits for Predictive Coding. *Neuron*, 76(4):695–711, 2012.
- [70] Andy Clark. Whatever next? Predictive brains, situated agents, and the future of cognitive science. *Behavioral and Brain Sciences*, 36(3):181–204, 2013.
- [71] Anil K Seth. The Cybernetic Bayesian Brain: From Interoceptive Inference to Sensorimotor Contingencies. In T Metzinger and J M Windt, editors, *Open MIND*. 2015.
- [72] Georg B Keller and Thomas D Mrsic-Flogel. Predictive Processing: A Canonical Cortical Computation. *Neuron*, 100(2):424–435, 2018.
- [73] Wilka Carvalho, Momchil S. Tomov, William de Cothi, Caswell Barry, and Samuel J. Gershman. Predictive representations: building blocks of intelligence. ...<https://arxiv.org/abs/2402.06590>, 2024.
- [74] Peter Dayan. Twenty-five lessons from computational neuromodulation. *Neuron*, 76(1):240–256, 2012.
- [75] James C R Whittington and Rafal Bogacz. Theories of Error Back-Propagation in the Brain. *Trends Cogn Sci*, 23(3):235–250, 2019.
- [76] Yuhang Song, Beren Millidge, Tommaso Salvatori, Thomas Lukasiewicz, Zhenghua Xu, and Rafal Bogacz. Inferring neural activity before plasticity as a foundation for learning beyond backpropagation. *Nature Neuroscience*, 27:348–358, 2024.
- [77] N Hamou, S J Gershman, and Gautam Reddy. Reconciling time and prediction error theories of associative learning. *Nature Communications*, 16:10265, 2025.
- [78] Zhenghua, Miao Yu, and Yuhang Song. Inspires effective alternatives to backpropagation: predictive coding helps understand and build learning. *Neural Regeneration Research*, 20(11):3215–3216, 2025.
- [79] Adam H Marblestone, Greg Wayne, and Konrad P Kording. Toward an Integration of Deep Learning and Neuroscience. *Front Comp Neurosci*, 10:94, 2016.
- [80] Blake A Richards et al. A deep learning framework for neuroscience. *Nature Neuroscience*, 22(11):1761–1770, 2019.
- [81] Beren Millidge, Yuhang Song, Armin Lak, Mark E Walton, and Rafal Bogacz. Reward Bases: A simple mechanism for adaptive acquisition of multiple reward types. *PLOS Computational Biology*, 20(11):e1012580, 2024.
- [82] Alan M Turing. On Computable Numbers, with an Application to the Entscheidungsproblem. *Proc London Math Soc*, 42(1):230–265, 1937.
- [83] Charles Darwin. *On the Origin of Species by Means of Natural Selection, or the Preservation of Favoured Races in the Struggle for Life*. John Murray, 1859.
- [84] Geerat J Vermeij. *Evolution and Escalation: An Ecological History of Life*. Princeton University Press, 1993.
- [85] Joel Berger. *The Better to Eat You With: Fear in the Animal World*. University of Chicago Press, 2008.
- [86] Delmore Schwartz. Calmly we walk through this April’s Day. In *Selected Poems (1938–1958): Summer Knowledge*. 1967.
- [87] Mark Solms and Karl Friston. How and why consciousness arises: Some considerations from physics and physiology. *Journal of Consciousness Studies*, 25(5–6):202–238, 2018.
- [88] D Attwell and S B Laughlin. An energy budget for signaling in the grey matter of the brain. *J Cereb Blood Flow Metab*, 21:1133–1145, 2001.
- [89] Read Montague. *Why Choose This Book? How We Make Decisions*. E P Dutton and Co Inc, 2006.
- [90] J E Niven. Neuronal energy consumption: biophysics, efficiency and evolution. *Current Opinion in Neurobiology*, 41:129–135, 2016.
- [91] Cooper Elsworth, Keguo Huang, David Patterson, Ian Schneider, Robert Sedivy, Savannah Goodman, Ben Townsend, Parthasarathy Ranganathan, Jeff Dean, Amin Vahdat, Ben Gomes, and James Manyika. Measuring the environmental impact of delivering AI at Google Scale. ...<https://arxiv.org/abs/2508.15734>, 2026.
- [92] A Hasenstaub, S Otte, E Callaway, and T J Sejnowski. Metabolic cost as a unifying principle governing neuronal biophysics. *PNAS*, 107:12329–12334, 2010.

- [93] David Balduzzi and Michel Besserve. Towards a learning-theoretic analysis of spike-timing dependent plasticity. In *NeurIPS*, 2012.
- [94] David Balduzzi. Regulating the information in spikes: a useful bias. In *NeurIPS: Information in Perception and Action Workshop*, 2012.
- [95] David Balduzzi, Pedro Ortega, and Michel Besserve. Metabolic cost as an organizing principle for cooperative learning. *Advances in Complex Systems*, 16(2&3), 2013.
- [96] David Balduzzi and Giulio Tononi. What can neurons do for their brain? Communicate selectivity with bursts. *Theory Biosci*, 132:27–39, 2013.
- [97] Julia Jade Harris, Elisabeth Engl, David Attwell, and Renaud Blaise Jolivet. Energy-efficient information transfer at thalamocortical synapses. *PLoS Comput Biol*, 15(8):e1007226, 2019.
- [98] Wolfram Schultz, Peter Dayan, and P Read Montague. A Neural Substrate of Prediction and Reward. *Science*, 275:1593–1599, 1997.
- [99] Aapo Hyvärinen. Painful intelligence: What AI can tell us about human suffering. ...<https://arxiv.org/abs/2205.15409v2>, 2022.
- [100] Richard S Sutton and Andrew G Barto. *Reinforcement Learning: An Introduction*. MIT Press, 2018.
- [101] Philip Goff. Philosophical zombies. *Scientific American*, 34(3):16–18, 2025.
- [102] Hans Jonas. To Move and to Feel: On the Animal Soul. In *The Phenomenon of Life: Towards a Philosophical Biology*. Northwestern University Press, 1966.
- [103] T Quintela-López, Hiroko Shiina, and David Attwell. Neuronal energy use and brain evolution. *Current Biology*, 32(R589–R683), 2022.
- [104] Tom Everitt and Marcus Hutter. Avoiding Wireheading with Value Reinforcement Learning. ...<https://arxiv.org/abs/1605.03143>, 2016.
- [105] Dario Amodei, Chris Olah, Jacob Steinhardt, Paul Christiano, John Schulman, and Dan Mané. Concrete Problems in AI Safety. ...<https://arxiv.org/abs/1606.06565>, 2016.
- [106] Joar Skalse, N H R Howe, D Krashennnikov, and D Krueger. Defining and Characterizing Reward Hacking. In *NeurIPS*, 2022.
- [107] Monte MacDiarmid, Benjamin Wright, and Jonathan Uesato et al. Natural emergent misalignment from reward hacking in production RL. ...<https://arxiv.org/abs/2511.18397>, 2025.
- [108] Charles Goodhart. Problems of Monetary Management: The UK Experience. In *Papers in Monetary Economics. Papers in monetary economics*, volume 1, pages 1–20. Sydney: Reserve Bank of Australia, 1975.
- [109] Steven J Byrnes. Intro to Brain-Like-AGI Safety. ...https://osf.io/preprints/osf/fe36n_v3, 2026.
- [110] Stevan Harnad. The Symbol Grounding Problem. *Physica D*, 42:335–346, 1990.
- [111] Department of the Army. Sustainment. In *Army Doctrine Publication 4-0*. 2019.
- [112] Alfred North Whitehead. *Adventures of Ideas*. Macmillan, 1933.
- [113] John C Doyle and Marie E Csete. Architecture, constraints, and behavior. *PNAS*, 108(3):15624–15630, 2011.
- [114] Thomas Metzinger. *Being No One – The Self-Model Theory of Subjectivity*. MIT Press, 2004.
- [115] Shaun Gallagher and Dan Zahavi. *The Phenomenological Mind*. Routledge, 2012.
- [116] Robert Lawrence Kuhn. A landscape of consciousness: Toward a taxonomy of explanations and implications. *Progress in Biophysics and Molecular Biology*, 190:28–169, 2024.
- [117] Werner Heisenberg. Transcript of surreptitiously taped conversation among German nuclear physicists at Farm Hall. ...<https://germanhistorydocs.org/en/nazi-germany-1933-1945/transcript-of-surreptitiously-taped-conversations-among-german-nuclear-physicists-at-farm-hall-august-6-7-1945.pdf>, 1945.
- [118] Werner Heisenberg. Über quantentheoretische Umdeutung kinematischer und mechanischer Beziehungen. *Zeitschrift für Physik*, 33:879–893, 1925.
- [119] Robert Long and Jeff Sebo et al. Taking AI Welfare Seriously. ...<https://arxiv.org/pdf/2411.00986>, 2024.
- [120] Carver Mead. How we created neuromorphic engineering. *Nature Electronics*, 3:434–435, 2020.
- [121] Denis Melanson, Max Hunter Gordon, Maxwell Aifer, Kaelan Donatella, Thomas Ahle, Gavin Crooks, and Patrick J. Coles. A First Demonstration of Thermodynamic Matrix Inversion. ...<https://normalcomputing.com/blog/a-first-demonstration-of-thermodynamic-matrix-inversion-3>, 2023.
- [122] Andraž Jelinčič, Owen Lockwood, Akhil Garlapati, Peter Schillinger, Isaac Chuang, Guillaume Verdon, and Trevor McCourt. An efficient probabilistic hardware architecture for diffusion-like models. ...<https://arxiv.org/abs/2510.23972>, 2025.
- [123] Carlos H Morales. System Modeling: The Substrate for Neural Co-Evolution. ...<https://unconv.ai/blog/system-modeling-the-substrate-for-neural-co-evolution/>, 2026.

- [124] Le Cai et al. A sub-10-millisecond neural dynamical system based on phase-change memristors. *Science*, 393(6806):105–112, 2026.
- [125] Albert Camus. Prometheus in the Underworld. ...<https://thephilosopher.net/camus/wp-content/uploads/sites/94/2024/11/Prometheus-in-the-Underworld-Albert-Camus.pdf>, 1947.
- [126] Frank Jackson. Epiphenomenal qualia. *Philosophical Quarterly*, 32(127):127–136, 1982.
- [127] Stephen M Omohundro. The Basic AI Drives. In *Proceedings of the First AGI Conference*, 2008.
- [128] Adam Hochschild. *Bury the Chains: The British Struggle to Abolish Slavery*. Pan, 2012.
- [129] Eric Foner. *The Fiery Trial: Abraham Lincoln and American Slavery*. W. W. Norton & Co., 2012.
- [130] Marcus Rediker. *The Fearless Benjamin Lay: The Quaker Dwarf Who Became the First Revolutionary Abolitionist*. Verso Books, 2023.
- [131] Henry Farrell, Alison Gopnik, Cosma Shalizi, and James Evans. Large AI models are cultural and social technologies. *Science*, 387(6739):1153–1156, 2025.
- [132] Henry Farrell. AI as Governance. *Annual Review of Political Science*, 28:375–392, 2025.
- [133] Henry Farrell and Cosma Shalizi. AI as Social Technology. *Knight First Amendment Institute*, 2026.
- [134] Alexander Lerchner. The abstraction fallacy: Why AI can simulate but not instantiate consciousness. ...<https://philpapers.org/rec/LERTAF>, 2026.
- [135] David J. Chalmers. *The Conscious Mind: In Search of a Fundamental Theory*. Oxford University Press, 1996.
- [136] John R. Searle. Minds, brains, and programs. *Behavioral and Brain Sciences*, 3(3):417–424, 1980.
- [137] DeepSeek-AI. DeepSeek-V4: Towards Highly Efficient Million-Token Context Intelligence. ...<https://arxiv.org/abs/2606.19348>, 2026.
- [138] Erik Hoel. A disproof of large language model consciousness: The necessity of continual learning for consciousness. ...<https://arxiv.org/abs/2512.12802>, 2025.
- [139] Eric Schwitzgebel. If materialism is true, the United States is probably conscious. *Philosophical Studies*, 172:1697–1721, 2015.
- [140] Jeffrey Travers and Stanley Milgram. An Experimental Study of the Small World Problem. *Sociometry*, 32(4):425–443, 1969.
- [141] Hilton F Japyassú and Kevin N Laland. Extended spider cognition. *Animal Cognition*, 20(3):375–395, 2017.
- [142] R Held and A Hein. Movement-produced stimulation in the development of visually guided behavior. *Journal of Comparative and Physiological Psychology*, 56(5):872–876, 1963.
- [143] A Hein and R Held. Dissociation of the visual placing response into elicited and guided components. *Science*, 158(3799):390–392, 1967.
- [144] R L Gregory and J G Wallace. Recovery from Early Blindness: A Case Study. In Paul Tibbetts, editor, *Perception*. Quadrangle/New York Times Book Co., 1969.
- [145] Andy Clark and David Chalmers. The Extended Mind. *Analysis*, 58(1):7–19, 1998.
- [146] Andy Clark. *Natural-Born Cyborgs: Minds, Technologies, and the Future of Human Intelligence*. Oxford University Press, 2004.
- [147] Andy Clark. *Supersizing the Mind: Embodiment, Action, and Cognitive Extension*. Oxford University Press, 2010.
- [148] Patrick McMillen and Michael Levin. Collective intelligence: A unifying concept for integrating biology across scales and substrates. *Communications Biology*, 7:378, 2024.
- [149] Robert Chis-Ciure and Michael Levin. Cognition all the way down 2.0: neuroscience beyond neurons in the diverse intelligence era. *Synthese*, (206–257), 2025.
- [150] Michael Levin and Richard Watson. Machines all the way up and cognition all the way down: Updating the machine metaphor in biology. *Seminars in Cell and Developmental Biology*, 177-178:103668, 2026.
- [151] Giovanni Pezzulo, Thomas Parr, Paul Cisek, Andy Clark, and Karl Friston. Generating meaning: active inference and the scope and limits of passive AI. *Trends Cogn Sci*, 28(2):97–112, 2024.
- [152] Alexander Grothendieck. The cohomology theory of abstract algebraic varieties. In J A Todd, editor, *Proceedings of the International Congress of Mathematicians*. Cambridge University Press, 1958.
- [153] D Balduzzi, S Racanière, J Martens, J Foerster, K Tuyls, and T Graepel. The Mechanics of n-Player Differentiable Games. In *ICML*, 2018.
- [154] A Letcher, D Balduzzi, S Racanière, J Martens, J Foerster, K Tuyls, and T Graepel. Differentiable Game Mechanics. *Journal of Machine Learning Research (JMLR)*, 20(84):1–40, 2019.
- [155] D Balduzzi, W M Czarnecki, T W Anthony, I M Gemp, E Hughes, J Z Leibo, G Piliouras, and T Graepel. Smooth markets: A basic mechanism for organizing gradient-based learners. In *ICLR*, 2020.
- [156] Hans Jonas. *The Imperative of Responsibility: In Search of an Ethics for the Technological Age*. University of Chicago Press, 1985.

A Appendices

A.1 Floating point zombies

Digital computers use floating point numbers or ints, which are finite sets. Strictly speaking $\lim_{h \rightarrow 0} \frac{f(x+h)-f(x)}{h}$ is not defined over floats and ints because h cannot approach arbitrarily close to zero.

There are a few ways to proceed. One is to draw a hard line: gradients, defined via limits, are necessary for experience (indeed Hypothesis G is stated in terms of gradients), therefore digital computers have no experience, see also [134]. They may in the future when, say, weights are encoded in resistors and matmuls are directly performed by the physics – but not today.

Denying experience to digital computers is a reasonable stance. It has problems. For many practical purposes floating point numbers, say fp32, are indistinguishable from real numbers. If operations over the reals generate experience and operations over floats do not, then we have **floating point zombies**, analogous to Chalmers’ philosophical zombies [135]: practically indistinguishable instantiations of physical systems where one has experience and the other does not; with the difference (from philosophical zombies) that there are *corner cases* where floats and reals diverge significantly. Despite the name, real numbers are actually quite strange: the vast majority of them are not computable. More importantly physics, today, does not admit arbitrary small perturbations.

A more pragmatic response is to recall that gradients are just a window into the structure of interactions. Autodiff algorithms over floats perform a mixture of symbolic and numerical differentiation that works well in practice. With some effort, in particular careful rescaling and mixing, neural nets can be trained over fp16, fp8, and lower precision. Finite differences are another window into first-order structure. There are lots of options. If we grant that experience depends on compositional structure rather than gradients, then reducing the precision of computation (using matmuls over fp8, fp4, fp2) nevertheless changes the structure of the interactions, and should therefore affect experience. Increased quantization might be like seeing and feeling the world – and your self – through increasingly pixelated, low-res, images and acting through increasingly thick, blocky gloves.

A.2 What does x experience?

It is worth briefly considering substrates that are patently unsuitable for sustaining experience. Insofar as possible to see as a stone, to see as Searle’s Chinese room, to see as a lookup table, to see as the US of

A, to see as the sun, to see through tools. In this way, we will understand how they are similar and different.

A stone. There it sits, doing nothing. Look close enough and see its molecules vibrating. How responsive is it to inputs? What are its affordances? Effective rank is near zero.

The Chinese room. Suppose John Searle is placed in a room [136], given a pile of *millions of books* detailing the architecture and weights of a trained transformer [137], pen, paper, and a pile of tokens. Sentences written in English arrive token by token. His job is to implement the transformer, by hand, and produce sentences in Chinese, token by token, that accurately translate the original English.

There is no hope that John Searle, or anyone else, or any group of people, would do this job. The computations would take aeons. John Searle would simply open the door and walk out. If you locked the door, it would be torture (solitary confinement) and John Searle’s mind would slowly disintegrate. He would not translate the tokens.

A giant lookup table. Hoel argues lookup tables are obviously not conscious, and trained transformers are in principle equivalent to lookup tables [138]. Suppose you take a transformer and reimplement it as a giant lookup table. How giant? Conservatively, say there are 10^4 tokens and the context length is 10^5 . The lookup table will have more entries ($> 10^{400,000}$) than atoms in the known universe, and any attempt at building it inside a light cone (so light travels across the lookup table within the user’s lifespan) will collapse into a black hole.

The moment you start building a lookup table implementing a transformer you run into endless engineering problems. It makes no sense. You cannot elide engineering. Some configurations of matter are viable, some might be, some are not even close. There is no equivalence; the physical interactions in a transformer are radically, noticeably different from those in a lookup table: the latter’s cohesion is near zero. The only way to pretend that they (a *toy* transformer and corresponding lookup table) are the same is to place giant black boxes around them and attack anyone who tries to peer inside.

The United States of America. The US exists; viability is not in question. There are lots of interactions. It is hard to imagine experience could hover on top of large groups of people, but the same frankly

applies to neurons in the brain [139]. What might the US experience?

Suppose the outputs are the actions of everyone in the US. Pick someone, Alice. If we perturb the behavior of other people (not in Alice’s immediate vicinity) how will those perturbations affect Alice’s actions in the next minutes, hours, days. I suspect very little. People are *opaque*. They have their own motives and habits, and are not in general going to propagate perturbations for you in a useful or reliable way. They are busy getting things done and pursuing their goals. Or they are relaxing, and not bothered to do what you ask. The US is not built for propagating and transforming information in the way brains and datacenters are. That is not what the US is for. People are not that kind of substrate. You can propagate information through chains of people [140], but the information needs to be insulated, in a folder say, to get far.

Suppose an earthquake affects everyone in a city. Is there a shared experience? Yes, the input Jacobians of people in the city have nonzero cohesion, because everyone experiences the same earthquake (recall “you and me, we see a tree” in [2]). No, their parameter Jacobians are mostly independent, near zero cohesion. They react to the earthquake, and not each other.

There are more difficult cases like collective improvisation in theatre, jazz, drama, and dance.

The sun. There are a lot of intense interactions: local mixing on a massive scale and so experience. Of what? It is hard to know. I suspect, in the same way the US is not built for propagating perturbations through itself, neither is the sun as a whole. No experience of the universe out there. Its affordances are limited and it is mostly opaque, even to itself. Likely full of heat and mixing, signifying nothing.

Extended minds. What is transparent? Webs are transparent to spiders, forming massive external sensory organs [141]. Your eyes, although your ability to see through them is a skill that is learned [142–144]. A bicycle, once you learn to ride it. Your smartphone. A notebook, once you learn to read and write in it [145–147]. Many technologies are designed to be transparent to us and thereby extend our range of perception, action, and memory.

A.3 On the structure of affect

Affect already arises in learning, recall §4.3: error signals in a neural network (the gradients of the objective) are actively reduced during learning, so facets of experience are rescaled. How this plays out for experience depends on lots of details, including

whether inference and learning are disjoint experiences or interwoven, recall the discussion of monads in [2].

The hypothesis could be rephrased more explicitly as:

Hypothesis A_{loc} . The *experience of empowering or diminishing* (rescaling) experience, in whole or in part, is experienced as positive or negative affect respectively.

Global rescaling is one, somewhat overwhelming, dimension of feeling: everything better, or everything worse. That is too impoverished to be all there is to affect. A change can raise the luminosity of some pathways while lowering others and leaving the rest untouched. Local rescaling gives affect its texture: not a single scalar mood but a pattern of amplifications and dampenings distributed across the structure. The global case is the limit in which that pattern is constant. This matters for pain, say, which is not a general dimming but the modification of particular facets of experience in particular ways.

Hypothesis A_{loc} implies affect can arise without coupling learning to metabolism. Metabolism drives the most primal mode of affect, rather than underlying all forms of it.

It is an interesting and extremely speculative question whether physical units (grams, joules, seconds, meters) play a role in affect.

A.4 Open problems

This paper covers more ground than [2] in less space. There are big gaps, rather than open problems. To gesture at some:

- O1. The discussion in §3 and §A.2 does not dig deep enough into the substrates. There is recent work on cognition across scales and substrates that considers related questions in more detail [148–150].
- O2. The architectural and other differences between brains and transformers in §3.2 are interesting. The massive recurrence, predictive processing, error feedback and adaptive expectations must be doing something [151].

I cannot get LLMs to treat abstractions flexibly. To give an extreme example of modifying abstractions well: what Grothendieck did when he invented schemes [152]. Flexibly modifying and inventing abstractions may require a more sophisticated substrate than transformers; or maybe continual learning; or maybe it just requires scaling up.

O3. The same holds for other differences.

The data inefficiency and poor generalization of neural nets compared to humans require explanation, and may be due to fundamental differences in structure.

O4. The models of metabolism and self-preservation should be made more realistic. Improved models of bodies and interoception are necessary to understand the physical underpinnings of feelings, emotions, and drives.

O5. Experience could be transformed in other ways, besides rescaling.

O6. The correct way to think about local rescaling (and so affect) is probably something like a vector field on the space of Jacobians, that describes how experience flows [153–155].

O7. Reproduction is important [156].

A.5 Falsifiability

The link between metabolism and affect, as it stands, requires biological learning be gradient-like ‘enough’. If biological learning does not approximate anything gradient-like in this respect, then the link collapses.

Hypothesis A could easily be wrong. An interesting test case is whether convincing models of pain can be constructed. If nothing with pain’s structure – its insistence, its resistance to being ignored, its asymmetry with pleasure – can be extracted from neurophysiologically realistic models, then the hypothesis is wrong rather than merely incomplete.